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Lineage and Release Timing in Correlated Errors of Open-Weight Language Models: A Precision-Gated, Pre-Registered Study of 977 Models

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ABSTRACT Ensembles and review pipelines built from several open-weight language models assume that their members err somewhat independently. When two models share an ancestor, or were released at the same time, that assumption may fail. We ask which of the two is associated with models choosing the *same wrong answer*. We first show why the question cannot be answered by decomposing per-model accuracy into family and release-era variance components: a precision analysis of the fitted random-effects model shows that, with five to seven model families and at most fourteen release quarters, neither share can be estimated to better than about ± 18 percentage points (root-mean-square error), whatever the number of models. We therefore move to a pair-level design. Using item-level predictions for 977 models from the Open LLM Leaderboard (MMLU, 14,042 items), validated item by item against the leaderboard's stored correctness, we resolve each model's lineage root from Hugging Face `base_model` declarations and relate pairwise agreement on wrong answers to shared lineage and to the gap between upload months. The analysis, sample rules and a simulation-based power gate were fixed before outcomes were computed; three dated amendments are disclosed, and a post-hoc high-replication audit shows that the gate's cross-leakage criterion is narrowly unmet at the largest simulated release-time effect (10.8% against a 10% limit). In the pre-registered primary analysis (590 models, 173,755 pairs), sharing a lineage root is associated with a 13.3 percentage-point higher probability of choosing the same wrong option when both models err (95% CI 11.8–14.8; delete-one-root jackknife CI 10.8–15.8), against a mean of 42.3%. The estimate is 11.7–13.4 points across all twelve pre-registered analyses and stays between 10.2 and 14.2 points under root-level inference, alternative weighting, a position-adjusted outcome and exploratory size controls. Item-difficulty-aware null models absorb almost all of the average agreement but leave the lineage contrast unchanged. Unadjusted, same-month pairs agree 8.7 points more than pairs released a year or more apart, but this gradient is fully accounted for by the similarity of the two models' accuracies; net of accuracy, release-month proximity shows no association of consistent sign with the choice of wrong option: the same-month coefficient is +0.004 (CI -0.003 to $+0.011$) in the primary analysis, small and positive under some weightings, and negative (-0.015 , CI -0.027 to -0.002) among models scoring at least 30%; for which items fail together, a small positive association appears in the full samples but not among models above 30%. The associations are observational and describe a population dominated by community fine-tunes; they do not separate the pretraining era of foundation models from their lineage.

INDEX TERMS Large language models, correlated errors, model lineage, ensembles, dyadic regression, pre-registration, Open LLM Leaderboard

I. INTRODUCTION

WHEN several public language models are asked the same hard question, they often fail together, and often in the same way. Kim *et al.* [1] document this across hundreds of open and hosted models: when two models are both wrong, they choose the same wrong option far more often than chance. For anyone who combines models—a majority vote, a routing scheme, or a review step that escalates only when models disagree—this matters directly. Agreement between members is informative only if their errors are not shared [18], [20].

Two explanations for shared errors compete. *Lineage*: a model fine-tuned from another inherits its parameters and therefore, plausibly, its blind spots. *Era*: models built at the same time draw on the same data, recipes and fashions and may acquire the same mistakes without any shared ancestry. The two are entangled, because descendants are always released after their ancestors. Which one dominates has a practical consequence: if lineage dominates, combining models from different base models should reduce correlated error; if era dominates, it may help little.

This paper reports a study that answers a narrower, well-posed version of that question, and the methodological path that led to it. An earlier version of this work attempted to decompose per-model benchmark accuracy into family and release-era variance components, applied a design “gate” to sixteen evaluated models, and found that the gate failed. On re-examination that version had three defects: its gate tested the rank of a fixed-effects design that the fitted random-effects model does not use; the evaluation code that produced its per-question predictions recorded only the first answer option, making all item-level outputs constant; and the variance decomposition, even when identified, cannot be estimated precisely with the handful of families and quarters available. Section III documents the last point, which motivates the pair-level design used here. Appendix A records what changed from the earlier version.

A. RESEARCH QUESTIONS

- 1) **RQ1:** Can family and release-era variance shares of per-model accuracy be estimated precisely from realistic model populations?
- 2) **RQ2:** Among open-weight models, is sharing a lineage root associated with choosing the same wrong answer, at equal release-time gap?
- 3) **RQ3:** Is release-time proximity associated with choosing the same wrong answer, at equal lineage status?

B. CONTRIBUTIONS

- 1) **A precision analysis for the per-model design.** We identify the correct identifiability condition for the crossed random-effects model and show by expected-information and Monte Carlo analysis that family and era shares are imprecise with the number of families and quarters that real populations offer (Section III).

- 2) **A pre-registered pair-level design with a simulation gate.** Agreement on wrong answers is modelled at the level of model pairs, with lineage and release-month gap as pair covariates and dyadic cluster-robust inference. Sample rules were fixed by a simulation-based power and separation gate before outcomes were read (Section IV).
- 3) **A validated item-level dataset.** We extract item-level predictions for 977 leaderboard models, validate every item against the leaderboard’s stored correctness and reproduce official aggregate scores, and resolve lineage from model-card declarations with a separately reported, lower-confidence extension (Section V).
- 4) **Results with explicit robustness.** Shared lineage is strongly and consistently associated with shared wrong answers; release-month proximity is not, in a consistent direction (Section VI).

Fig. 1 summarizes the study. Steps 2–6 were fixed in a dated pre-registration before any pair outcome was computed; the audits in step 7 are partly post hoc and are labelled as such where they are reported. Throughout, “pre-registered” refers to that document, “exploratory” to analyses specified after it, and “post hoc” to analyses added after the results were known.

II. RELATED WORK

Correlated errors in LLMs. Kim *et al.* [1] measure pairwise agreement on wrong answers across 349 Hugging Face models and 71 HELM models and regress it on pair characteristics, finding that a shared provider, a shared base architecture and similar size predict higher agreement, and that larger and more accurate models have more correlated errors even across architectures and providers. They suggest that newer models may be converging in their outputs but do not test this over release time. Chen [2] shows, across 67 models, that the gain from routing, voting or mixture-of-agents is capped by the rate at which all models fail the same query, and that average pairwise error correlation understates that rate. Kuai *et al.* [8] audit 18 models from six families for synchronized failures with difficulty-weighted dependence metrics, relate dependence to judges over-endorsing responses, and reweight verifier ensembles accordingly. Goel *et al.* [3] propose a chance-adjusted similarity metric based on overlapping mistakes and report that mistakes become more similar as capability increases. Our temporal predictor is different from both accuracy-related correlation and model age: it is the *proximity* of two models’ release months, estimated at equal lineage status and conditional on both models’ accuracies, which the model controls for (the accuracy-sum coefficient is positive, consistent with [1]). We make no claim about absolute model age.

Null models for excess agreement. Jo, Garg and Raghavan [23] argue that whether models agree “more than expected” depends on two analyst choices: the null model of independence, where accounting for item difficulty leads to very different conclusions from nulls that do not, and the pop-

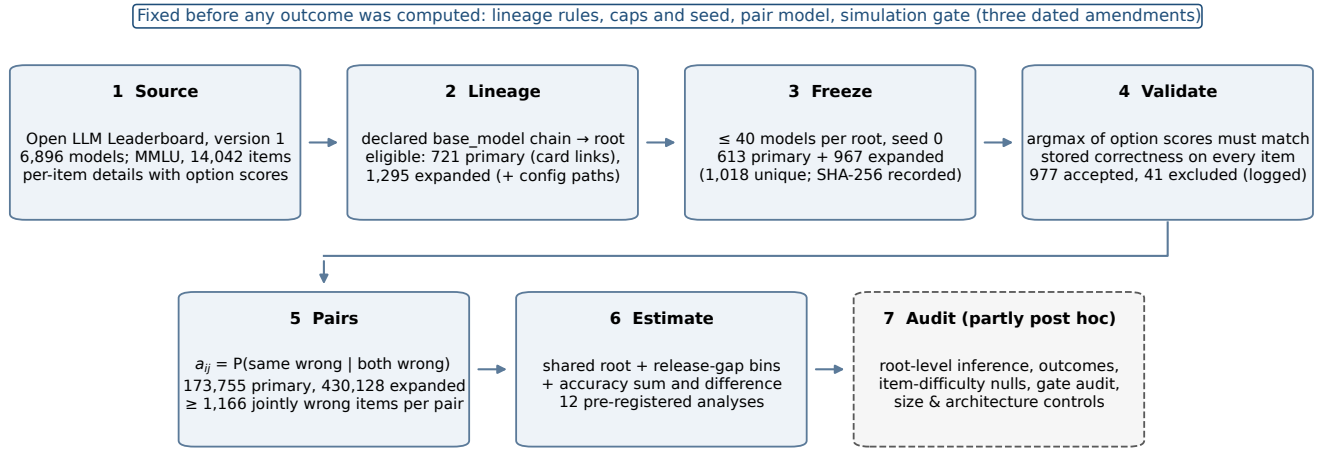


FIGURE 1: Study workflow. Counts are read from the analysis outputs. Steps 2–6 were fixed before outcomes were computed (three dated amendments, Section IV-C); step 7 combines pre-registered secondary analyses with exploratory and post-hoc audits. Primary and expanded populations differ only in the lineage evidence accepted (Section V-B); a_{ij} is the outcome of Eq. (2).

ulation of models and items compared. They conclude that monoculture is a context-dependent inference rather than an absolute property of models. This bears directly on our study, which uses one benchmark and one leaderboard population. Three estimands are in use: correlated correctness (whether two models fail the same items, e.g. the phi coefficient of error indicators, or co-failure in excess of an item-difficulty null); conditional agreement on the wrong option (whether two models that both fail an item choose the same distractor [1]); and accuracy-adjusted similarity [3]. Our primary estimand is the second and our secondary the first. More importantly, our quantity of interest is a *contrast*—the difference in agreement between pairs that share a lineage root and pairs that do not, at equal release-month gap—not the *level* of excess agreement that the null-model debate concerns. A null that shifts all pairs’ expected agreement equally leaves such a contrast unchanged; a null that differs between related and unrelated pairs does not. Section VI-D therefore re-estimates both outcomes against item-difficulty-aware nulls.

Our contribution. We do not claim to be the first study of correlated errors in language models. To our knowledge, and based on the work reviewed here, this is the first study to examine declared fine-tune ancestry and release-month proximity jointly as correlates of conditional same-wrong-answer agreement. Relative to the work above, the outcome is conditional same-wrong-option agreement between pairs of models (not the all-fail rate of an ensemble [2], judge dependence [8] or pipeline measurement error [13]); lineage is each model’s declared fine-tune ancestry resolved to a root checkpoint (not provider or architecture labels [1], family membership [8] or dataset lineage [7]); and release time enters as the proximity of two models’ upload months, estimated jointly with lineage. We separate the two explicitly: lineage is taken from declared ancestry rather than provider labels, release time is the gap between upload months in categorical

bins, inference accounts for the dependence of pairs that share a model or a lineage root, and the design was checked by a pre-specified simulation gate.

Lineage and ancestry. Laufer, Oderinwale and Kleinberg [4] map the family trees of about 1.86 million Hugging Face models from their declared parents and study how metadata traits—licenses, language support, model cards—are inherited and drift along lineages. Horwitz, Shul and Hoshen [5] recover model trees from model weights alone. Both treat lineage as the object of study; neither measures whether related models make the same errors on benchmark items. PhyloLM [6] reconstructs phylogenies of language models from output similarity; Li *et al.* [7] reconstruct the lineage of post-training *datasets* and show that benchmark contamination can propagate along it. These works infer or exploit lineage of models or data; we take lineage from declarations and ask how much error agreement it accounts for relative to release time.

Variance components and dyadic inference. Crossed variance-component models estimated by restricted maximum likelihood [9]–[11] underlie the per-model analysis in Section III; generalizability theory [12] asks the same question of measurement facets, and Messing [13] shows that ignoring design choices in LLM pipelines (judge model, temperature, prompt) makes evaluation uncertainty, including on MMLU, look smaller than it is. Pairwise outcomes are not independent, because every model appears in many pairs; we use dyadic cluster-robust variance estimators [14]–[16] and a delete-one-root jackknife [17].

Ensembles and monoculture. Ensemble gains depend on error diversity [18]–[20]; algorithmic monoculture arguments [21], [22] treat correlated error as an input. Our results bear on which kind of diversity is associated with less shared error, without testing ensembles directly.

III. WHY PER-MODEL VARIANCE DECOMPOSITION IS NOT ENOUGH

A. MODEL AND IDENTIFICATION

Let Y_i be a per-model trait (e.g., accuracy), with family $f(i)$ and release quarter $e(i)$:

$$Y_i = \mu + \alpha_{f(i)} + \beta_{e(i)} + u_i, \quad (1)$$

$$\alpha \sim \mathcal{N}(0, \sigma_L^2), \quad \beta \sim \mathcal{N}(0, \sigma_E^2), \quad u \sim \mathcal{N}(0, \sigma_U^2),$$

so that $\text{Var}(\mathbf{y}) = \sigma_L^2 \mathbf{Z}_F \mathbf{Z}_F^\top + \sigma_E^2 \mathbf{Z}_E \mathbf{Z}_E^\top + \sigma_U^2 \mathbf{I}$, with only the intercept as a fixed effect. The variance components are identified when the covariance basis $\{\mathbf{Z}_F \mathbf{Z}_F^\top, \mathbf{Z}_E \mathbf{Z}_E^\top, \mathbf{I}\}$ is linearly independent. The rank of the fixed-effects dummy design $[\mathbf{1} \mid \mathbf{Z}_F \mid \mathbf{Z}_E]$, used as the gate in the earlier version of this work, is not the relevant condition for this model: for the sixteen models evaluated there, that design is rank-deficient (rank 14 of 15) while the covariance basis has full rank (3 of 3). Its condition number also depends on the arbitrary choice of reference level (TABLE I).

B. PRECISION

Identification does not imply precision. For each design we compute the expected restricted-likelihood information $I_{jk} = \frac{1}{2} \text{tr}(PV_j PV_k)$ at four share scenarios (lineage-dominant 0.5/0.2/0.3, era-dominant 0.2/0.5/0.3, balanced, and low-signal 0.1/0.1/0.8), the delta-method standard error of each share, and, as the verdict, the Monte Carlo root-mean-square error (RMSE) of the share estimates under the same REML estimator (500 replicates per scenario). The analytic standard error tracks the Monte Carlo RMSE closely (e.g., 0.289 versus 0.285 for the sixteen-model design), so it can be used before any measurement.

No design reaches an RMSE of 0.10 for both shares (TABLE I): not the sixteen evaluated models (0.30), not the 47-model candidate frame (0.19), and not the 30-model configuration the earlier version described as sufficient (0.19). Removing the single model responsible for the fixed-effects rank deficiency leaves the RMSE unchanged (0.302). The constraint is the number of *levels* of each factor, not the number of models. In staggered designs (Fig. 2), bringing both shares within ± 0.10 requires on the order of sixty families and fourteen quarters; with eight quarters the era share never falls below about 0.14. Because release quarters are few by construction, a per-model decomposition is a poor instrument for RQ2–RQ3. This answers RQ1 in the negative and motivates the pair-level design.

IV. PAIR-LEVEL DESIGN

A. OUTCOME

For models i and j evaluated on the same $K = 14,042$ MMLU items [24], let c_{ik} be model i 's chosen option on item k (the argmax of its per-option log-likelihoods) and g_k the correct option. The outcome is the conditional agreement [1]

$$a_{ij} = \frac{\sum_k \mathbf{1}[c_{ik} = c_{jk} \neq g_k]}{\sum_k \mathbf{1}[c_{ik} \neq g_k] \mathbf{1}[c_{jk} \neq g_k]}, \quad (2)$$

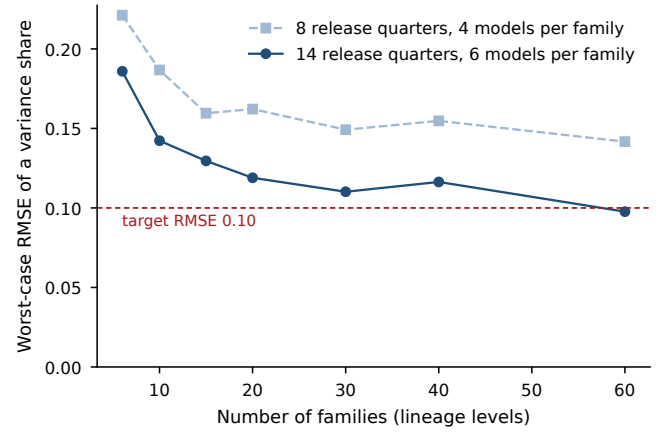


FIGURE 2: Precision of per-model variance shares versus the number of families. Worst-case Monte Carlo RMSE of the family or era share across four scenarios (300 replicates each) for staggered designs with 8 or 14 release quarters. The dashed line marks an RMSE of 0.10; only the largest design with 14 quarters (60 families, 360 models) reaches it.

the probability of choosing the same wrong option given that both are wrong. Every pair contributes all items on which both models are wrong; in the data this denominator ranges from 1,166 to 10,820 items (median 4,109 in the primary sample), so every a_{ij} is estimated from more than a thousand items.

What the outcome measures. Each MMLU item has three wrong options. If two models that both err chose among them independently and uniformly, a_{ij} would be $1/3$; values above $1/3$ arise either because some distractors are more attractive to all models, or because the two models share a preference. The outcome therefore separates two questions that are often merged: *whether* two models fail on the same items (correlated correctness, measured here by the secondary outcome, the phi coefficient of the two error indicators), and *how* they fail when they do (which distractor they choose). Conditioning on joint failure removes most of the mechanical dependence of agreement on accuracy, because two weak models are not counted as agreeing merely for being wrong often; it does not remove all of it, which is why accuracy enters the model below. Because a_{ij} is a ratio over a pair-specific set of items, pairs with low accuracy are evaluated on more, and on average easier, items than pairs with high accuracy; the item-difficulty-aware nulls of Section VI-D address this directly.

Estimand. The quantity of interest is a *contrast* between kinds of pairs, not the level of a_{ij} : how much more often two models agree on wrong options when they descend from the same root, at equal release-month gap and equal accuracies, and how much more often they agree when released close together, at equal lineage status and equal accuracies. Fig. 3 illustrates the two predictors. Every pair has one value of each, so the design crosses them: same-root pairs occur at several gaps and same-month pairs occur with and without a shared root. The two terms are separated only to the extent that these combinations are populated, which Section V-B

TABLE I: Precision of family and era variance shares for per-model traits. FE: fixed-effects dummy design used as a gate in the earlier version; κ range is over every choice of reference levels. Covariance-basis rank 3/3 means the variance components are identified. The last column is the worst-case Monte Carlo RMSE of either share across four scenarios (500 replicates each).

Design	Families	Quarters	FE full rank	$\kappa(X^\top X)$ range	Cov. basis	Worst RMSE
16 evaluated models	5	11	no	singular	3/3	0.302
22-model selection	6	14	yes	285–1526	3/3	0.256
47-model candidate frame	6	14	yes	166–1116	3/3	0.189
30-model “sufficient” design	6	8	yes	91–144	3/3	0.194
42-model sweep design	7	8	yes	83–113	3/3	0.182
nested reference (72)	6	6	no	singular	2/3	not identified

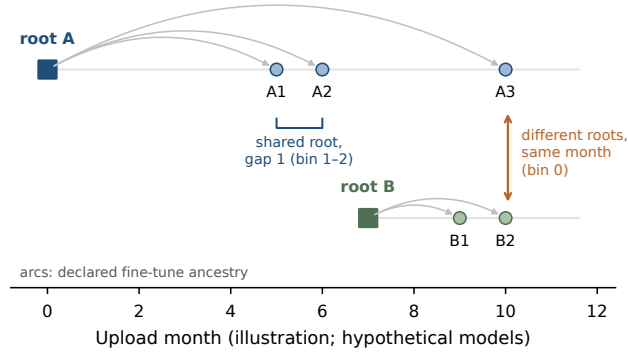


FIGURE 3: The two pair-level predictors (illustration with hypothetical models). Squares are lineage roots and circles fine-tunes placed at their upload month; arcs show declared ancestry. A1 and A2 share a root and were uploaded one month apart (shared root, gap bin 1–2); A3 and B2 descend from different roots but were uploaded in the same month (no shared root, gap bin 0). Every pair carries one value of each predictor, and the pair model estimates each association at equal levels of the other.

documents.

B. MODEL AND INFERENCE

$$a_{ij} = b_0 + b_L s_{ij} + \sum_g b_{E,g} \mathbf{1}[|m_i - m_j| \in g] + c_1(\text{acc}_i + \text{acc}_j) + c_2|\text{acc}_i - \text{acc}_j| + \varepsilon_{ij}, \quad (3)$$

where s_{ij} indicates a shared lineage root, m_i is the model’s upload month, and the gap bins are 0, 1–2, 3–5 and 6–11 months with 12 or more as reference. b_L is the association of shared lineage with agreement at equal time gap; $b_{E,0}$ is the association of same-month release at equal lineage status. Accuracy terms control for the fact that weaker models have more, and different, errors. Because every model enters many pairs, pair residuals are dependent; the pre-registered standard errors are dyadic cluster-robust [14]–[16], allowing arbitrary correlation between any two pairs that share a model. With $\mathbf{x}_p$ the covariate vector and $\hat{\varepsilon}_p$ the residual of pair p , the estimator is

$$\hat{V}(\hat{\mathbf{b}}) = (X^\top X)^{-1} \left(\sum_{p,q: p \cap q \neq \emptyset} \hat{\varepsilon}_p \hat{\varepsilon}_q \mathbf{x}_p \mathbf{x}_q^\top \right) (X^\top X)^{-1}, \quad (4)$$

where $p \cap q \neq \emptyset$ means that pairs p and q share at least one model (including $p = q$). Section VI-B adds two procedures that treat the lineage root as the unit of dependence: root-level dyadic clustering, Eq. (4) with “share a model” replaced

by “touch a common root”, and a delete-one-root jackknife, which refits the model G times, each time omitting every pair that involves one of the G roots, and estimates the variance as $\frac{G-1}{G} \sum_g (\hat{\mathbf{b}}_{(g)} - \bar{\mathbf{b}})^2$. Root-level procedures matter because fine-tunes of one root may share errors beyond what their shared membership in pairs captures.

Why accuracy is controlled. Two models of similar accuracy tend to fail on similar items, and the items that strong models fail are the hardest ones, which may have more attractive distractors. Agreement can therefore vary with accuracy for reasons unrelated to lineage or release time. The sum and absolute difference of the two accuracies were pre-registered as controls. They are not neutral: if release time shifts accuracy, part of a release-time relationship runs through accuracy, and the adjusted coefficient describes only the remainder. We therefore also report the unadjusted model (Section VI).

C. SIMULATION GATE AND AMENDMENTS

Before outcomes were read, the design was checked by simulation on the actual roster (roots and months only). For model i with ability $\theta_i \sim \mathcal{N}(0.3, 1)$ and item k with difficulty $d_k \sim \mathcal{N}(0, 1)$, the model answers correctly with probability $\text{logit}^{-1}(\theta_i - d_k)$. If it errs, its wrong option is

$$c_{ik} = \begin{cases} R_{r(i),k} & \text{with probability } \lambda_L, \\ E_{m(i),k} & \text{with probability } \lambda_E, \\ G_k & \text{with probability } \lambda_G = 0.15, \\ \text{uniform} & \text{otherwise,} \end{cases} \quad (5)$$

where $R_{r,k}$ is a wrong option preferred by root r on item k , $E_{m,k}$ a wrong option preferred in month m (persisting from one month to the next with probability 0.8, so that adjacent months share preferences), and G_k a wrong option preferred by all models. Under this model a lineage effect and a release-time effect act through the same mechanism, a shared preferred distractor, which is the scenario in which the two are hardest to tell apart. The full pair pipeline, Eq. (2)–(3) with dyadic standard errors, was applied to each simulated data set. Three properties were checked: null rejection when $\lambda_L = \lambda_E = 0$; *cross-leakage*, the rate at which the wrong term rejects (the lineage term under a release-time-only truth, or a release-month term under a lineage-only truth), which measures whether the confounding of lineage and time is handled; and power. The pre-registered pass criteria were: null rejection at most 10%; cross-leakage (the lineage term

TABLE II: Pair-level simulation gate (primary roster, 14,042 simulated items, 20 replicates per truth). Null: largest rejection rate under no effect. Leakage: largest cross-term rejection. Power: rejection of the lineage / era term at the stated effect weight.

Cap	N	Null	Leakage	Power $\lambda=0.05$	Power $\lambda=0.03$
15	561	15%	20%	100% / 100%	50% / 75%
40	613	5%	10%	100% / 100%	40% / 95%
none	721	10%	15%	100% / 100%	95% / 95%

rejecting under era-only truth, or vice versa) at most 10%; and power at least 80% for $\lambda = 0.05$ on both terms. TABLE II shows the gate on the primary roster for three caps on the number of models per lineage root.

Post-hoc audit of the gate. With 20 replicates per truth, the rates in TABLE II move in steps of five points. After the results were known, and in response to review, we re-ran the same simulation with 500 replicates per truth (TABLE III and Fig. 4, Wilson 95% intervals). This audit does not alter the dated decision above, and we report it against the original criteria. For the chosen design (primary, cap 40), the null rejection rate is 5.8–7.8% (upper limits up to 10.5%) and power at $\lambda = 0.05$ is 100% for both terms, so those criteria are met. Power at $\lambda = 0.03$ is 63% for lineage and 88% for release month; the gate did not require power at this effect size. The leakage criterion is *not* met at the strongest era effect: with $\lambda_E = 0.10$ the lineage term rejects in 10.8% of replicates (95% CI 8.4–13.8; 12.6% in the expanded sample), against a threshold of 10%. Every other leakage rate (era into lineage at $\lambda_E \leq 0.05$, and lineage into either release-month term at all three λ_L) lies between 3.6 and 7.8%. The leak is a small positive bias—the lineage estimate rises by 1.4×10^{-4} , about 2% of the simulated era effect—arising because pairs sharing a root were released closer together; applied to the largest observed release-month coefficient ($|b_{E,0}| \leq 0.015$), it implies a bias of about 3×10^{-4} on a lineage estimate of 0.133. The audit also shows that in the primary design the dyadic standard errors are about 4–6% smaller than the simulated sampling spread, consistent with null rejection rates slightly above 5%; in the expanded design they are calibrated. The gate therefore passes its null and power criteria and narrowly fails its leakage criterion at the largest simulated era effect; the root-level inference procedures in Section VI-B, which give wider intervals, are the appropriate protection. The audit also does not support the 20-replicate choice among caps: the uncapped design (7.2% leakage) and a cap of 15 (7.2%) meet all three criteria, and only the chosen cap of 40 narrowly fails; the earlier ranking reflected simulation noise. We keep the dated choice. Because the cap-15 sample is a subset of the cap-40 sample under the same seeded ordering, we re-estimated the primary model on it (exploratory; 540 validated models): the shared-root coefficient is +0.146 (jackknife SE 0.015) and the same-month coefficient +0.005 (SE 0.005). The uncapped design could not be analysed, because only the capped samples were downloaded.

Three amendments were made and dated in the project’s

pre-registration document, each before the quantities it concerns were computed: (1) the primary data source moved from the second to the first version of the leaderboard, whose item-level files are not access-gated; (2) the per-root cap was set to 40, the only setting meeting all gate criteria (TABLE II), and the gate was re-run with the real item count after an initial run with 1,500 items understated power; (3) after the per-model accuracies of the first 443 validated files showed many near-chance models, two sensitivity analyses were added before any pair outcome was computed: S1 adds the total-variation distance $\frac{1}{2} \sum_o |\pi_i(o) - \pi_j(o)|$ between the two models’ distributions π over chosen answer options as a covariate (an answer-option-distribution control: two models that both favor the same letters agree more for that reason alone, whatever their lineage), and S2 restricts to models with accuracy ≥ 0.30 .

V. DATA

A. SOURCE AND VALIDATION

The first version of the Open LLM Leaderboard [25] evaluated thousands of open-weight models on MMLU (5-shot) with the LM Evaluation Harness [27] and published per-item details, including per-option log-likelihoods. For each model we select the most complete run covering all 57 subjects and read four fields per item: the per-option log-likelihoods, the gold option, the leaderboard’s stored per-item correctness and, where present, an item hash. We accept the model only if (i) all 14,042 items are present, (ii) the argmax of the per-option scores reproduces the stored per-item correctness on *every* item, and (iii) item order and gold answers match a reference model. Three storage layouts occur; in the newest, the gold column is empty and gold is taken from the reference by position, which check (ii) then validates (a misaligned gold vector fails on roughly three quarters of items). Extracted macro-accuracy equals the leaderboard’s official results files for the models we spot-checked (43.80, 70.06 and 75.56). For each accepted model we store the chosen option (the argmax), the gold option, the item identifier and hash, the run identifier and the file layout; the per-option scores are used for validation and not retained. Fig. 5 shows the procedure and where models were rejected.

Check (ii) is the substantive one. The earlier version of this work failed precisely here: its evaluation code read one field per item and recorded the first answer option for every question, producing constant predictions that no aggregate statistic flagged (Appendix A). Requiring the extracted answer to reproduce the leaderboard’s own correctness flag on all 14,042 items rules out that failure, a misaligned item order, and a misread gold column, because each of these produces thousands of disagreements. It cannot detect an error in the leaderboard’s evaluation itself, such as an unsuitable prompt format for a particular model (Section VIII).

B. LINEAGE AND RELEASE TIME

Lineage is resolved by following each model’s declared `base_model` on Hugging Face to a root, collapsing repository

TABLE III: Post-hoc gate audit (500 replicates per truth; Wilson 95% intervals, in percent). Null: largest rejection rate of any term under no effect. Leakage: largest cross-term rejection across $\lambda \in \{0.03, 0.05, 0.10\}$. Power: rejection of the lineage (L) or release-month (E) term. Not pre-registered.

Sample	Cap	N	Null	Leakage	L, $\lambda=0.05$	E, $\lambda=0.05$	L, $\lambda=0.03$	E, $\lambda=0.03$
Primary	15	561	6.0 [4.2, 8.4]	7.2 [5.2, 9.8]	99.6 [98.6, 99.9]	100.0 [99.2, 100.0]	40.2 [36.0, 44.6]	83.8 [80.3, 86.8]
Primary	40	613	7.8 [5.8, 10.5]	10.8 [8.4, 13.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	63.2 [58.9, 67.3]	87.8 [84.6, 90.4]
Primary	none	721	6.6 [4.7, 9.1]	7.2 [5.2, 9.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	90.4 [87.5, 92.7]	90.2 [87.3, 92.5]
Expanded	40	967	5.6 [3.9, 8.0]	12.6 [10.0, 15.8]	100.0 [99.2, 100.0]	100.0 [99.2, 100.0]	93.6 [91.1, 95.4]	97.8 [96.1, 98.8]

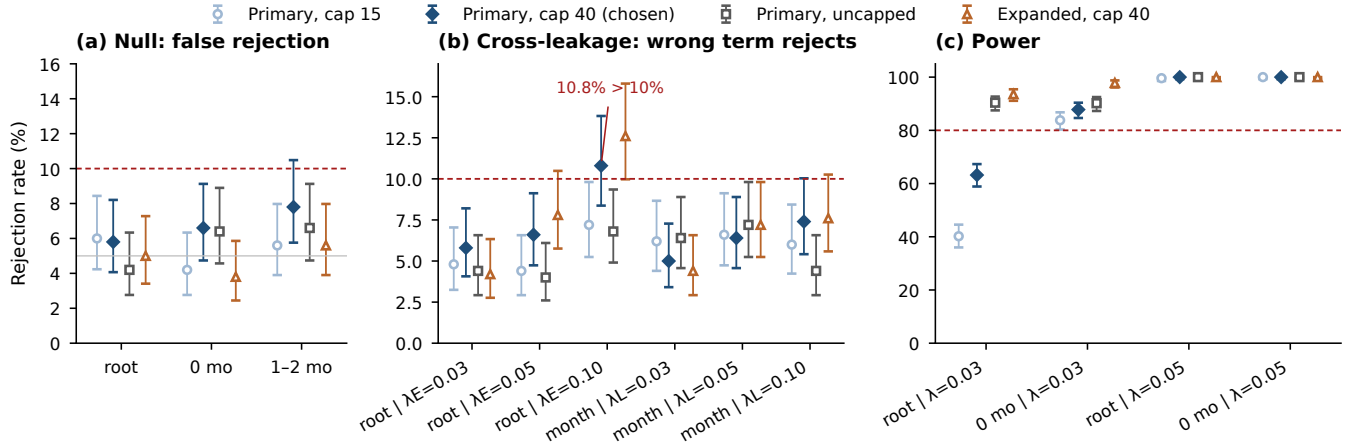


FIGURE 4: Post-hoc audit of the simulation gate (500 replicates per truth; bars are Wilson 95% intervals). (a) Rejection rate of each term when no effect is simulated. (b) Cross-leakage: rejection of the lineage term under release-time-only truths (λ_E), and of the more often rejecting of the 0- and 1–2-month terms under lineage-only truths (λ_L). (c) Power of the lineage and same-month terms. Dashed lines mark the pre-registered criteria (10% for null and leakage, 80% for power at $\lambda = 0.05$; no power criterion applied at $\lambda = 0.03$). The chosen design (primary, cap 40) exceeds the leakage criterion at $\lambda_E = 0.10$ (10.8%, interval 8.4–13.8%), as does the expanded design (12.6%). Not pre-registered.

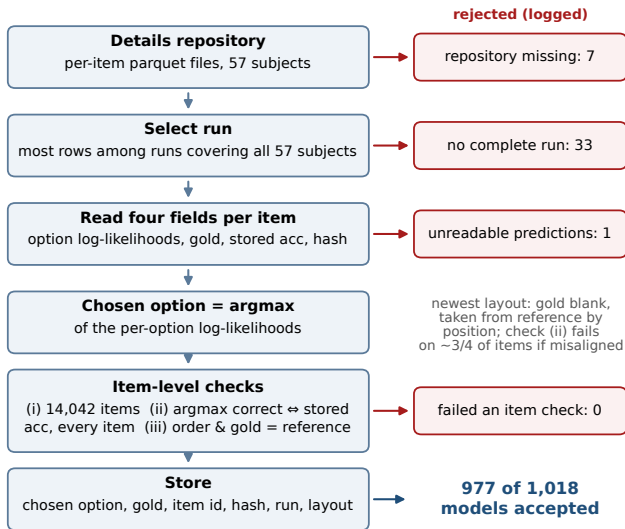


FIGURE 5: Prediction extraction and validation for the 1,018 models in the two frozen samples. Rejections are logged with their reason and were not replaced. Check (ii) compares the argmax of the per-option log-likelihoods with the leaderboard's stored per-item correctness on every item; no model that reached this step failed it.

renames and a short, documented list of straight re-uploads and quantisations. A chain is unresolved if it passes through a merge (several parents), reaches a deleted or invalid ancestor, or ends at a fine-tune with no declared parent. Two populations are frozen: *primary*, in which every link comes from a model card, and *expanded*, which additionally accepts a model's own first link from the `_name_or_path` field of its configuration file when that field names an existing Hugging Face repository (valid for 1,066 of 2,978 models lacking a card declaration, of which 574 then resolved to an eligible root; local paths, merge-tool paths and self-references were rejected). A *strict* variant keeps only chains ending at a checkpoint the leaderboard lists as pretrained. Release time is the Hub creation month of the model repository (27 distinct months, March 2022 to July 2024).

C. MODEL FLOW

Fig. 6 reconciles the populations. The two frozen samples (at most 40 models per root, fixed seed) overlap in 562 models and together comprise 1,018; 977 passed validation and 41 were excluded and documented (33 without a complete MMLU run, 7 whose details repository is missing, and 1 with unreadable predictions). No excluded model was replaced. Every exclusion was decided before any pair outcome was computed.

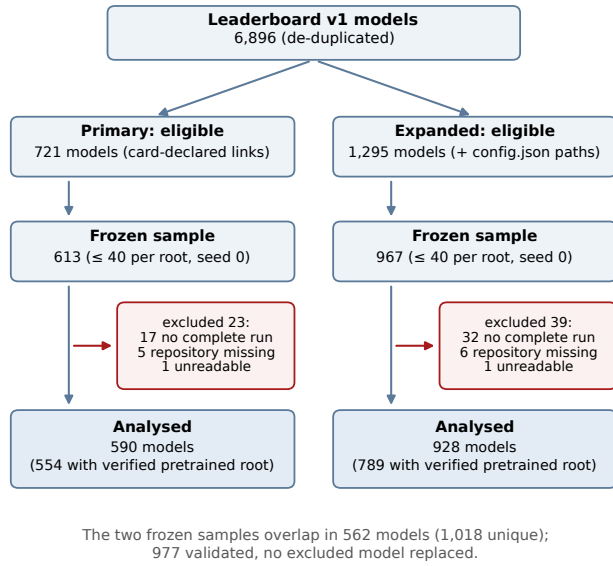


FIGURE 6: Model populations and exclusions. The primary population accepts only lineage links declared on model cards; the expanded population also accepts a model’s first link from its configuration file. Each frozen sample holds at most 40 models per lineage root (seed 0); exclusions are counted per sample, so models in both samples appear in both columns. “Verified pretrained root”: the strict variant used in two pre-registered analyses per population.

The analysed primary sample comprises 590 models from 351 lineage roots, of which 63 roots contribute two or more models; the expanded sample comprises 928 models from 422 roots (108 with two or more). The largest roots are Llama-3-8B and Mistral-7B-v0.1 (39 and 37 models in the primary sample). Median accuracy is 0.415; 247 of 590 primary models score below 0.30, close to chance for four options, which motivated S1 and S2.

How lineage and release time overlap. Fig. 7 places the primary models on a grid of lineage root by upload month. Two features matter for interpretation. First, most roots contribute a single model (288 of 351 in the primary sample); such models enter only different-root pairs, and the lineage contrast is identified by the 63 roots with two or more models. Second, the fine-tunes of a root are uploaded within a few months of one another, and the fine-tunes of the largest roots were uploaded in late 2023 and 2024, so same-root pairs are concentrated at short gaps (711, 1,004, 259, 45 and 1 pairs in the 0, 1–2, 3–5, 6–11 and 12+ month bins). Release-month gaps of a year or more are populated almost entirely by different-root pairs. The lineage coefficient is therefore estimated mainly from comparisons at gaps under six months, and the release-time coefficients mainly from different-root pairs; the simulation gate was designed to test whether this structure lets either term absorb the other.

VI. RESULTS

A. PRE-REGISTERED ANALYSES

The mean agreement is 0.423 (primary). Kim *et al.* report the same mean for their Hugging Face population [1], but that population comes from a later leaderboard snapshot with a different question set (12,032 questions), so the coincidence is not evidence of validity. Unadjusted, pairs sharing a root agree on 73.1% of jointly wrong items and other pairs on 42.0% (Fig. 9). TABLE IV and Fig. 8 report all twelve pre-registered analyses.

Fig. 9 shows the raw pattern behind these estimates. At every release-month gap at which both kinds of pair occur, same-root pairs agree on 71–75% of jointly wrong items and different-root pairs on 40–45%. Among different-root pairs agreement declines slowly with the gap, from 0.45 in the same month to 0.37 at a year or more.

Lineage (RQ2). In the primary analysis, sharing a root is associated with a 13.3-point higher agreement (95% CI 11.8–14.8). Across the twelve analyses the estimate ranges from 11.7 to 13.4 points; every confidence interval excludes zero by a wide margin. Controlling for the answer-option distribution (S1) leaves it unchanged; restricting to models above 0.30 accuracy (S2) reduces it slightly, to about 12 points.

Release time (RQ3). In the primary analysis the same-month coefficient is +0.004 (CI −0.003 to +0.011, $p = 0.26$). It is positive and nominally significant when the answer-option distribution is controlled (+0.010, CI +0.0005 to +0.0195, $p = 0.04$), and negative when near-chance models are excluded (−0.015, CI −0.027 to −0.002, $p = 0.02$ in the primary S2 analysis; −0.009, $p = 0.07$ in the expanded S2 analysis). The 1–2, 3–5 and 6–11-month bins show the same pattern. The data therefore do not show a release-time association of consistent sign; where coefficients are nominally significant they are an order of magnitude smaller than the lineage association and change sign between specifications. This is a lack of consistent evidence, not evidence that the association is zero. It is also conditional on accuracy. TABLE V reports the model with and without the pre-registered accuracy terms. Without them, release proximity has a graded association (same month +0.087, falling to +0.029 at 6–11 months) and the shared-root coefficient is +0.283; with them, every release-month coefficient is near zero and the shared-root coefficient is +0.133. The unadjusted release-time gradient, and about half of the unadjusted lineage gap, are therefore accounted for by models of similar accuracy agreeing more [1]. The two columns answer different questions. The adjusted coefficient is the association of release proximity *conditional on* both models’ accuracies; because release time can itself shift accuracy, conditioning may remove part of a genuine release-time relationship, and the raw coefficient is the better description of how much more contemporaneous models agree overall. Fig. 10 shows the comparison for both populations.

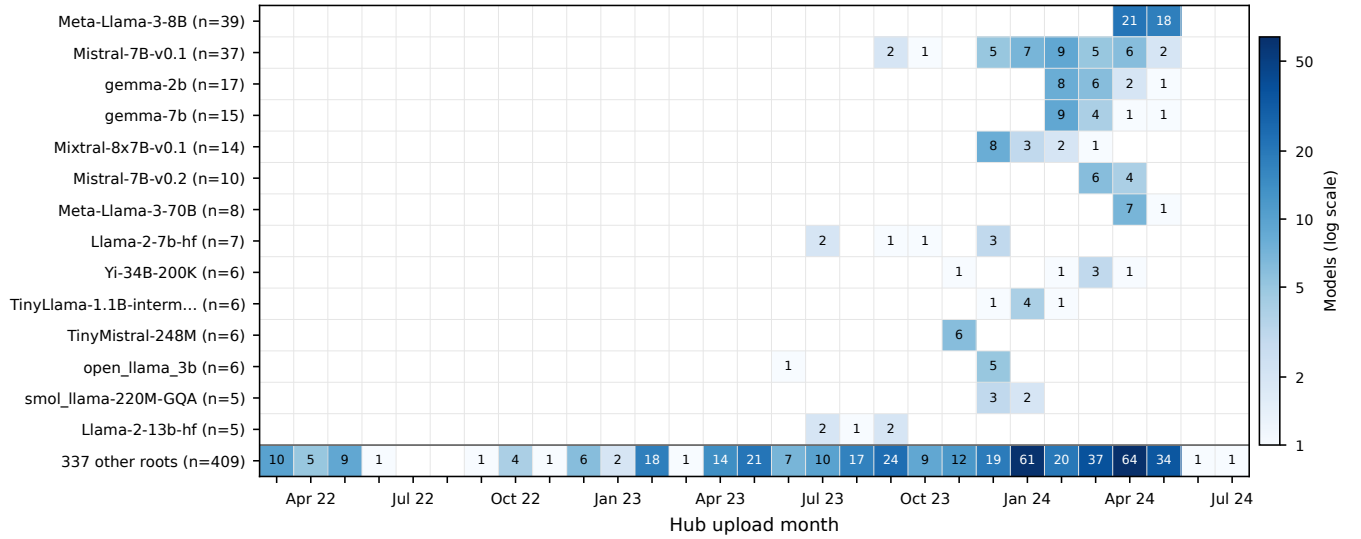


FIGURE 7: Lineage roots by upload month (primary sample, 590 models). Each cell counts the analysed models of one root uploaded in one month (log color scale; empty cells blank). The 14 largest roots are shown individually; the bottom row pools the remaining 337 roots, most of which contribute a single model. Fine-tunes of a root cluster within a few months, so same-root pairs are concentrated at short release-month gaps.

TABLE IV: Pre-registered analyses. Coefficients of Eq. (3) with 95% confidence intervals from dyadic (model-level) cluster-robust standard errors. “Same month” is the 0-month gap bin relative to 12 or more months; p refers to it.

Analysis	Models	Pairs	Shared root	Same month	p (same month)
Primary	590	173,755	+0.133 [+0.118, +0.148]	+0.0039 [-0.0029, +0.0106]	0.263
Primary, verified roots	554	153,181	+0.134 [+0.118, +0.149]	+0.0034 [-0.0032, +0.0100]	0.313
Primary + S1 (option distribution)	590	173,755	+0.133 [+0.118, +0.148]	+0.0100 [+0.0005, +0.0195]	0.039
Primary, S2 (acc ≥ 0.30)	343	58,653	+0.120 [+0.105, +0.135]	-0.0145 [-0.0268, -0.0022]	0.021
Primary + S1, S2	343	58,653	+0.117 [+0.102, +0.131]	-0.0124 [-0.0323, +0.0076]	0.225
Primary, verified + S1, S2	319	50,721	+0.118 [+0.103, +0.132]	-0.0138 [-0.0336, +0.0059]	0.170
Expanded	928	430,128	+0.127 [+0.115, +0.140]	+0.0059 [-0.0003, +0.0121]	0.061
Expanded, verified roots	789	310,866	+0.131 [+0.118, +0.144]	+0.0047 [-0.0016, +0.0109]	0.145
Expanded + S1 (option distribution)	928	430,128	+0.126 [+0.114, +0.138]	+0.0136 [+0.0048, +0.0223]	0.002
Expanded, S2 (acc ≥ 0.30)	589	173,166	+0.122 [+0.111, +0.133]	-0.0089 [-0.0185, +0.0008]	0.071
Expanded + S1, S2	589	173,166	+0.120 [+0.110, +0.130]	-0.0030 [-0.0172, +0.0113]	0.683
Expanded, verified + S1, S2	481	115,440	+0.125 [+0.115, +0.136]	-0.0110 [-0.0261, +0.0042]	0.157

TABLE V: Raw and accuracy-adjusted associations (primary and expanded samples; 95% CIs from the delete-one-root jackknife). Raw: Eq. (3) without the accuracy terms. Adjusted: the pre-registered model. Release-month bins are relative to 12 or more months.

Term	Primary		Expanded	
	Raw	Adjusted	Raw	Adjusted
Shared root	+0.283 [+0.251, +0.315]	+0.133 [+0.108, +0.158]	+0.240 [+0.205, +0.275]	+0.127 [+0.110, +0.145]
Same month	+0.087 [+0.051, +0.122]	+0.004 [-0.005, +0.013]	+0.092 [+0.065, +0.119]	+0.006 [-0.002, +0.014]
1–2 months	+0.085 [+0.048, +0.123]	-0.000 [-0.007, +0.007]	+0.087 [+0.060, +0.114]	+0.002 [-0.005, +0.010]
3–5 months	+0.065 [+0.031, +0.099]	-0.001 [-0.009, +0.007]	+0.078 [+0.050, +0.107]	+0.001 [-0.007, +0.008]
6–11 months	+0.029 [+0.012, +0.046]	-0.001 [-0.008, +0.006]	+0.055 [+0.031, +0.079]	+0.002 [-0.006, +0.009]

B. DEPENDENCE AND INFERENCE

The 173,755 primary pairs are generated by 590 models, and pairs within a lineage root may be dependent beyond shared membership. TABLE VI compares the pre-registered model-level dyadic standard errors with root-level dyadic clustering, which allows correlation between any two pairs touching the same root, and a delete-one-root jackknife, which removes all pairs involving one root at a time. Root-level procedures widen the interval of the lineage coefficient, by 10–26% with

root-level dyadic clustering and by 39–63% with the jackknife; its intervals remain far from zero (primary: jackknife CI 10.8–15.8). For the same-month coefficient in the primary S2 analysis, the jackknife interval includes zero ($p = 0.07$), so that negative estimate is not robust to root-level dependence. Fig. 11 compares the three procedures. The jackknife gives the widest interval for both terms in all four samples, so it is the conservative reference. Whichever procedure is used, every shared-root interval lies above 0.09, and of the twelve

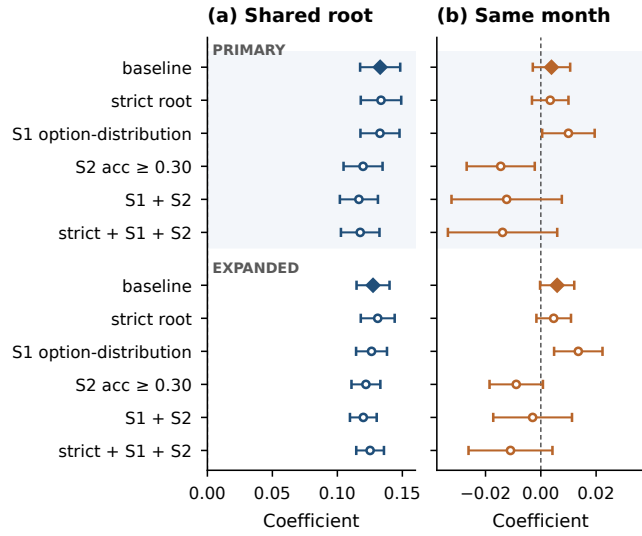


FIGURE 8: Shared-root and same-month coefficients across the twelve pre-registered analyses (95% CIs from model-level dyadic cluster-robust standard errors). Diamonds: the primary specification of each population. Strict: root verified as a pretrained checkpoint. S1: adds the answer-option-distribution covariate. S2: models with accuracy ≥ 0.30 . Note the different horizontal scales: every shared-root interval lies above 0.10, while same-month estimates lie within ± 0.02 and change sign between specifications. Root-level intervals are in Fig. 11.

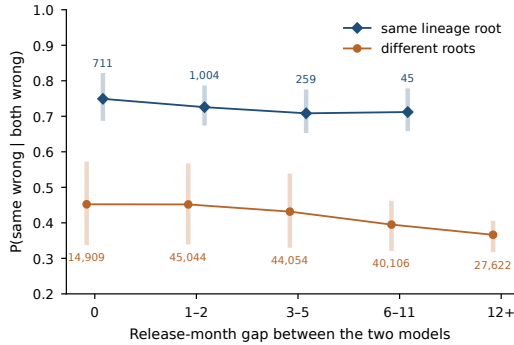


FIGURE 9: Unadjusted agreement on wrong answers by release-month gap for pairs sharing a lineage root and other pairs (primary sample). Points: mean of a_{ij} ; shaded bars: interquartile range across pairs; labels: number of pairs. Bins with fewer than 20 pairs are omitted (one same-root pair is 12 or more months apart). The gap between the two lines is present at every release-month gap; the decline of the lower line with gap is the unadjusted release-time gradient that accuracy similarity accounts for (Fig. 10).

same-month intervals shown only the model-level and root-level dyadic intervals of the primary S2 analysis exclude zero.

C. OUTCOME DEFINITION AND WEIGHTING

TABLE VII varies the outcome and weighting. Weighting pairs by their number of jointly wrong items raises the lineage estimate slightly (14.2 points) and makes the same-month coefficient positive ($+0.012$, $p = 0.02$). The pre-registered secondary outcome, the phi correlation of error indicators, measures whether two models fail the *same items* rather than

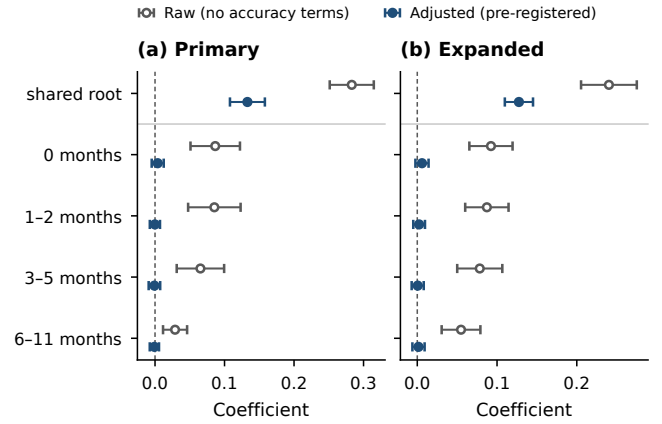


FIGURE 10: Raw and accuracy-adjusted associations (delete-one-root jackknife 95% CIs). Raw: the pre-registered model without the two accuracy terms. Adjusted: the pre-registered model. Release-month bins are relative to a gap of 12 or more months. Adding the accuracy terms halves the shared-root coefficient and removes the graded release-time association in both populations. Values: TABLE V.

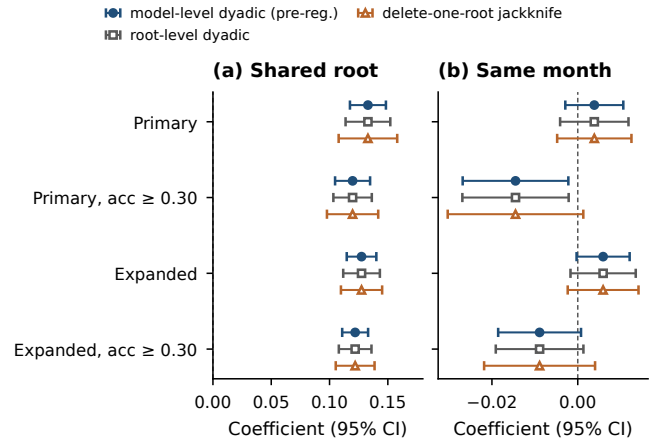
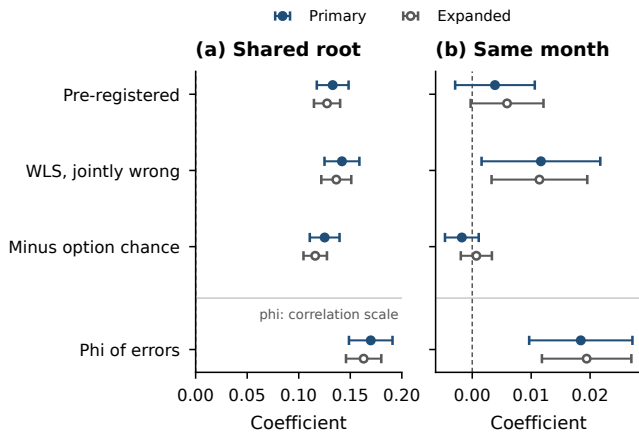


FIGURE 11: Sensitivity of the intervals to the unit of dependence. The pre-registered model-level dyadic intervals allow correlation between pairs sharing a model; root-level dyadic intervals allow it between pairs touching the same lineage root; the jackknife removes one root, and every pair involving it, at a time. Values: TABLE VI.

choose the same wrong option; it shows a lineage association of $+0.170$ and a smaller, positive same-month association ($+0.018$, CI $+0.010$ to $+0.027$). Subtracting the agreement expected from each model's marginal distribution over answer options (exploratory; an item-level version follows in Section VI-D) leaves the lineage estimate at 12.5 points and the same-month coefficient at -0.002 . Restricting to pairs with at least 1,000 jointly wrong items changes nothing, because every pair already exceeds that threshold. A small, positive association between release proximity and *which items* fail together is therefore plausible; an association with *which wrong option* is chosen is not supported once the answer-option distribution is accounted for. Fig. 12 shows that

TABLE VI: Inference audit. [†]Pre-registered. Root-level procedures treat lineage roots as the unit of dependence.

Sample	Term	Standard error	Estimate [95% CI]	<i>p</i>
primary	same root	model-dyadic [†]	+0.133 [+0.118, +0.148]	<0.001
primary	same root	root-dyadic	+0.133 [+0.114, +0.152]	<0.001
primary	same root	delete-one-root jackknife	+0.133 [+0.108, +0.158]	<0.001
primary	same month	model-dyadic [†]	+0.0039 [-0.0029, +0.0106]	0.263
primary	same month	root-dyadic	+0.0039 [-0.0041, +0.0118]	0.343
primary	same month	delete-one-root jackknife	+0.0039 [-0.0048, +0.0125]	0.383
primary, S2	same root	model-dyadic [†]	+0.120 [+0.105, +0.135]	<0.001
primary, S2	same root	root-dyadic	+0.120 [+0.103, +0.136]	<0.001
primary, S2	same root	delete-one-root jackknife	+0.120 [+0.098, +0.142]	<0.001
primary, S2	same month	model-dyadic [†]	-0.0145 [-0.0268, -0.0022]	0.021
primary, S2	same month	root-dyadic	-0.0145 [-0.0269, -0.0021]	0.022
primary, S2	same month	delete-one-root jackknife	-0.0145 [-0.0303, +0.0013]	0.072
expanded	same root	model-dyadic [†]	+0.127 [+0.115, +0.140]	<0.001
expanded	same root	root-dyadic	+0.127 [+0.112, +0.143]	<0.001
expanded	same root	delete-one-root jackknife	+0.127 [+0.110, +0.145]	<0.001
expanded	same month	model-dyadic [†]	+0.0059 [-0.0003, +0.0121]	0.061
expanded	same month	root-dyadic	+0.0059 [-0.0017, +0.0135]	0.128
expanded	same month	delete-one-root jackknife	+0.0059 [-0.0023, +0.0142]	0.160
expanded, S2	same root	model-dyadic [†]	+0.122 [+0.111, +0.133]	<0.001
expanded, S2	same root	root-dyadic	+0.122 [+0.108, +0.136]	<0.001
expanded, S2	same root	delete-one-root jackknife	+0.122 [+0.105, +0.139]	<0.001
expanded, S2	same month	model-dyadic [†]	-0.0089 [-0.0185, +0.0008]	0.071
expanded, S2	same month	root-dyadic	-0.0089 [-0.0191, +0.0013]	0.088
expanded, S2	same month	delete-one-root jackknife	-0.0089 [-0.0218, +0.0040]	0.178

**FIGURE 12: Outcome and weighting variants** (95% CIs, model-level dyadic). WLS: pairs weighted by the number of jointly wrong items. Minus option chance: agreement minus the agreement expected from the two models' marginal distributions over answer options (exploratory). Phi: the pre-registered secondary outcome, a correlation of error indicators on a different scale. The variant restricted to pairs with at least 1,000 jointly wrong items is identical to the pre-registered fit and is omitted. Values: TABLE VII.

the lineage estimate is stable across these definitions, while the same-month estimate is sensitive to both weighting and outcome.

D. ITEM-DIFFICULTY-AWARE NULLS

Added after the results were known, in response to review and following [23], these analyses are exploratory. Two nulls condition on the item. N1 sets the expected agreement on each item to $q_k = \sum_o p_k(o)^2$, where $p_k(o)$ is the share of

wrong models choosing wrong option o on item k , so that an item with one attractive distractor is expected to produce agreement regardless of lineage; the outcome is observed agreement minus the mean of q_k over the pair's jointly wrong items. N1a computes p_k from all models, which is conservative because a large lineage cluster pulls p_k towards its own choices; N1b weights every lineage root equally. N2 fits a Rasch model, $\text{logit } P(\text{wrong}_{ik}) = d_k - \theta_i$, by joint maximum likelihood and measures co-failure in excess of $\sum_k \hat{P}_{ik} \hat{P}_{jk}$, per item. TABLE VIII reports both with delete-one-root jackknife intervals.

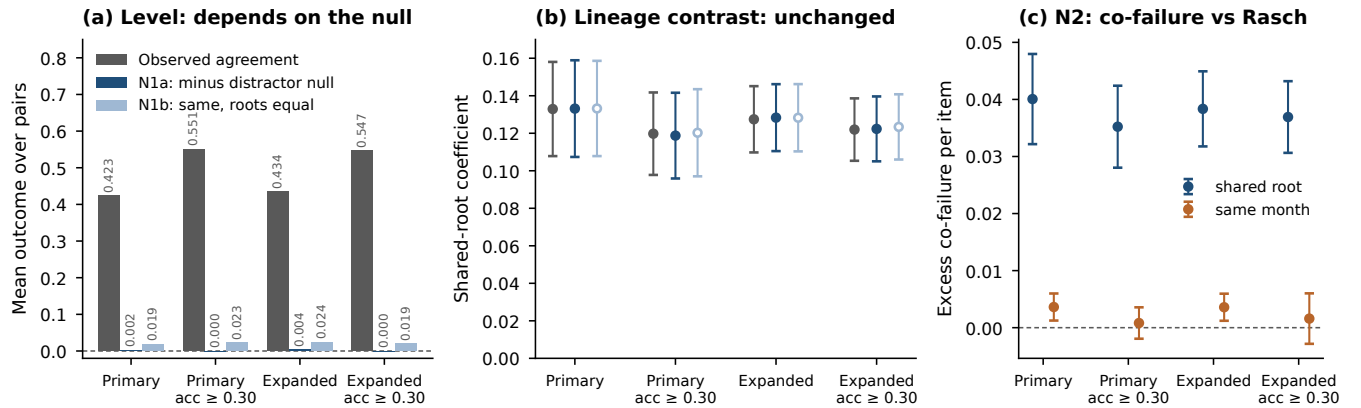
Item-level distractor popularity accounts for almost all of the average agreement: the mean excess falls from 0.423 to 0.002 (N1a) and 0.019 (N1b), consistent with the observation of [23] that levels of excess agreement depend on the null. The lineage contrast does not change: pairs sharing a root exceed the item-aware expectation by 13.3 points (N1a and N1b, primary), and co-fail on 4.0% more items than the Rasch model predicts. The same-month coefficient for the choice of wrong option is unchanged (+0.004, jackknife $p = 0.37$). For co-failure, a small positive same-month association remains in the full samples (+0.0036 per item, jackknife $p = 0.003$) but not among models scoring at least 30% ($p = 0.56$ and 0.48). Fig. 13 shows the central point: the item-level nulls move the level of the outcome from 0.42–0.55 to between 0.000 and 0.024, yet the shared-root contrast changes by less than 0.002 in every sample. The nulls therefore shift related and unrelated pairs almost equally.

E. EXPLORATORY CONTROLS

Two checks were not pre-registered (TABLE IX). Adding model size and same-architecture controls reduces the lin-

TABLE VII: Outcome and weighting variants (dyadic cluster-robust 95% CIs). Same month: 0-month gap bin relative to 12 or more months.

Sample	Variant	Shared root	Same month	<i>p</i> (same month)
Primary	primary outcome, OLS (pre-registered)	+0.133 [+0.118, +0.148]	+0.0039 [-0.0029, +0.0106]	0.263
Primary	WLS, weight = jointly-wrong items	+0.142 [+0.125, +0.159]	+0.0117 [+0.0016, +0.0217]	0.023
Primary	pairs with ≥ 1000 jointly-wrong items	+0.133 [+0.118, +0.148]	+0.0039 [-0.0029, +0.0106]	0.263
Primary	phi of error indicators (secondary)	+0.170 [+0.149, +0.191]	+0.0184 [+0.0097, +0.0272]	<0.001
Primary	agreement minus option-distribution chance (exploratory)	+0.125 [+0.111, +0.140]	-0.0018 [-0.0046, +0.0011]	0.227
Expanded	primary outcome, OLS (pre-registered)	+0.127 [+0.115, +0.140]	+0.0059 [-0.0003, +0.0121]	0.061
Expanded	WLS, weight = jointly-wrong items	+0.136 [+0.122, +0.151]	+0.0114 [+0.0033, +0.0195]	0.006
Expanded	pairs with ≥ 1000 jointly-wrong items	+0.127 [+0.115, +0.140]	+0.0059 [-0.0003, +0.0121]	0.061
Expanded	phi of error indicators (secondary)	+0.163 [+0.146, +0.180]	+0.0194 [+0.0118, +0.0270]	<0.001
Expanded	agreement minus option-distribution chance (exploratory)	+0.116 [+0.105, +0.127]	+0.0007 [-0.0019, +0.0033]	0.604

**FIGURE 13: Item-difficulty-aware nulls** (exploratory, added after review; delete-one-root jackknife 95% CIs). (a) Mean outcome over pairs: observed agreement on wrong answers, and agreement in excess of the item-level distractor null computed from all models (N1a) or with roots weighted equally (N1b). (b) Shared-root coefficient for the same three outcomes. (c) Excess co-failure over a Rasch null (N2), a measure of whether the same items fail, per item: shared-root and same-month coefficients. Values: TABLE VIII.**TABLE VIII: Item-difficulty-aware nulls** (exploratory, added after review). Mean: average outcome across pairs. Estimates with 95% CIs and *p*-values from the delete-one-root jackknife.

Sample	Outcome	Mean	Shared root	<i>p</i>	Same month	<i>p</i>
Primary	Same-wrong agreement (primary outcome)	0.423	+0.133 [+0.108, +0.158]	<0.001	+0.0039 [-0.0048, +0.0125]	0.383
Primary	N1a: minus item distractor null	0.002	+0.133 [+0.107, +0.159]	<0.001	+0.0040 [-0.0047, +0.0127]	0.367
Primary	N1b: as N1a, roots weighted equally	0.019	+0.133 [+0.108, +0.159]	<0.001	+0.0038 [-0.0049, +0.0124]	0.393
Primary	N2: co-failure over Rasch null (per item)	-0.000	+0.040 [+0.032, +0.048]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Primary, s2	Same-wrong agreement (primary outcome)	0.551	+0.120 [+0.098, +0.142]	<0.001	-0.0145 [-0.0303, +0.0013]	0.072
Primary, s2	N1a: minus item distractor null	-0.000	+0.119 [+0.096, +0.142]	<0.001	-0.0138 [-0.0310, +0.0034]	0.116
Primary, s2	N1b: as N1a, roots weighted equally	0.023	+0.120 [+0.097, +0.143]	<0.001	-0.0143 [-0.0315, +0.0030]	0.104
Primary, s2	N2: co-failure over Rasch null (per item)	-0.000	+0.035 [+0.028, +0.042]	<0.001	+0.0008 [-0.0019, +0.0036]	0.561
Expanded	Same-wrong agreement (primary outcome)	0.434	+0.127 [+0.110, +0.145]	<0.001	+0.0059 [-0.0023, +0.0142]	0.160
Expanded	N1a: minus item distractor null	0.004	+0.128 [+0.110, +0.146]	<0.001	+0.0061 [-0.0024, +0.0146]	0.160
Expanded	N1b: as N1a, roots weighted equally	0.024	+0.128 [+0.110, +0.146]	<0.001	+0.0062 [-0.0023, +0.0146]	0.154
Expanded	N2: co-failure over Rasch null (per item)	-0.000	+0.038 [+0.032, +0.045]	<0.001	+0.0036 [+0.0012, +0.0060]	0.003
Expanded, s2	Same-wrong agreement (primary outcome)	0.547	+0.122 [+0.105, +0.139]	<0.001	-0.0089 [-0.0218, +0.0040]	0.178
Expanded, s2	N1a: minus item distractor null	-0.000	+0.122 [+0.105, +0.140]	<0.001	-0.0088 [-0.0226, +0.0051]	0.214
Expanded, s2	N1b: as N1a, roots weighted equally	0.019	+0.123 [+0.106, +0.141]	<0.001	-0.0088 [-0.0229, +0.0052]	0.216
Expanded, s2	N2: co-failure over Rasch null (per item)	-0.000	+0.037 [+0.031, +0.043]	<0.001	+0.0016 [-0.0028, +0.0060]	0.480

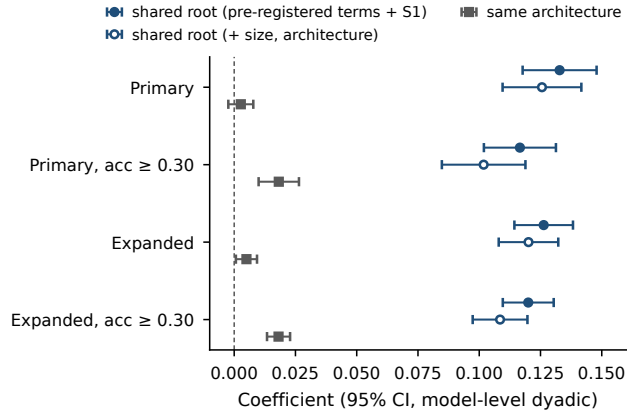
age estimate to 10.2–12.6 points; a shared architecture is associated with a further 0.3–1.8 points. Shared lineage and shared architecture are nested in this population (fine-tunes keep their base architecture), so these coefficients partition one association rather than test two independent ones. Fig. 14 shows the comparison. The reduction is largest among models scoring at least 30%, where a shared architecture also carries its largest coefficient (about 0.018 in both populations); in the full samples the same-architecture coefficient is 0.003–0.005.

VII. DISCUSSION

What the results say. Within this population, which open-weight model a fine-tune descends from is strongly associated with which wrong answers it gives: pairs sharing a root agree on wrong answers about 13 points more often, a 31% increase over the mean. The time at which a model was uploaded adds little, and nothing of consistent sign, to the choice of wrong option. This is consistent with Kim *et al.*'s finding that a shared provider and architecture predict agreement [1], and

TABLE IX: Exploratory controls (not pre-registered; all include the S1 covariate). Entries: estimate (standard error).

Sample	Models	Shared root (root-clustered SE)	Shared root (+ size, architecture)	Same month (+ size, architecture)	Same architecture
Primary	590	+0.1328 (0.0096)	+0.1256 (0.0082)	+0.0071 (0.0050)	+0.0027 (0.0026)
Primary, S2	343	+0.1166 (0.0086)	+0.1018 (0.0087)	-0.0205 (0.0110)	+0.0182 (0.0042)
Expanded	928	+0.1263 (0.0072)	+0.1201 (0.0062)	+0.0115 (0.0045)	+0.0050 (0.0022)
Expanded, S2	589	+0.1200 (0.0069)	+0.1085 (0.0057)	-0.0088 (0.0081)	+0.0181 (0.0024)

**FIGURE 14: Size and architecture controls** (exploratory; 95% CIs from model-level dyadic standard errors). Both models include the pre-registered terms and the S1 covariate; the second adds the absolute difference and sum of log parameter counts and a same-architecture indicator. Values: TABLE IX.

refines it: with lineage taken from declared ancestry and time measured directly, the association runs through lineage.

What they do not say. The association is observational. Fine-tunes of one base share its parameters, its pretraining data and often similar post-training data; the design does not separate these routes, and it does not show that fine-tuning causes inherited errors. Nor does it show that release time is irrelevant: the pretraining era of a foundation model is a property of its root and therefore nested in lineage here; the release-time estimate is net of accuracy, and contemporaneous models agree more partly because they are similarly accurate; and the phi analysis suggests a small association between release proximity and which items fail together.

Adjusted and unadjusted associations. Without the accuracy terms, models uploaded in the same month agree on wrong answers 8.7 points more often than models uploaded a year or more apart; with them, the difference is 0.4 points. Which number is relevant depends on the role accuracy plays. If release time affects agreement partly *through* accuracy—for example, because models of one period share a level of capability and therefore fail on the same hard items—then accuracy is a mediator, the unadjusted gradient is the total association, and the adjusted coefficient is the part that does not run through accuracy. If instead accuracy and release time are both consequences of something else, such as which base models were available, adjustment removes a confounded component. The data cannot distinguish these readings. What they do show is narrower: once two models' accuracies are

known, also knowing how far apart they were released adds nothing of consistent sign about whether they will choose the same distractor, whereas knowing whether they share a root still adds about 13 points. For lineage the choice matters less, because the shared-root coefficient is large under every specification, falling from 28.3 points unadjusted to 13.3 points adjusted.

Size of the lineage association. In raw terms, when two fine-tunes of the same root both fail an item they choose the same distractor about three times in four (73.1%), against about two times in five (42.0%) for models from different roots. Net of accuracy and release gap the difference is 13.3 points. The item-difficulty nulls show that this is not an artefact of particular items attracting all models to one distractor: relative to an item-level expectation the excess for same-root pairs is again 13.3 points, while the excess for an average pair is close to zero. Related models are, in this sense, the main source of agreement on wrong answers beyond what the items themselves induce.

Implications. For practitioners combining open-weight models, the results suggest that drawing members from different base models is associated with less shared wrong-answer behavior than drawing them from one base at different times. We did not evaluate ensembles directly, and the effect on ensemble accuracy depends on how members are combined [2].

VIII. LIMITATIONS

- 1) **Population.** The leaderboard population is self-selected and dominated by community fine-tunes of a few dozen bases between 2022 and 2024. Results may not hold for frontier or closed models, or for separately pretrained models released at the same time.
- 2) **Era definition.** Release time is the Hub creation month of the fine-tune, not the time its data were collected; pretraining era cannot be separated from lineage in this design.
- 3) **Lineage measurement.** Declared `base_model` fields can be wrong or missing; models with undeclared lineage are excluded from the primary population, and the expanded population relies on configuration paths that are weaker evidence. Conclusions agree across primary, expanded and strict populations.
- 4) **Benchmark.** One benchmark (MMLU, four options) is used. Many models are near chance; the S1 and S2 analyses and the item-level nulls address this, but preferences over answer options remain a concern for any agreement measure on multiple-choice tests.

- 5) **Simulation gate.** The pre-registered gate used 20 replicates per truth. A 500-replicate audit confirms its null and power criteria but finds cross-leakage of 10.8–12.6% at the largest simulated release-time effect, above the 10% criterion, and model-level standard errors about 4–6% too small in the primary design; we therefore report root-level intervals alongside the pre-registered ones.
- 6) **Evaluation pipeline.** All predictions were produced by the leaderboard’s evaluation pipeline (the LM Evaluation Harness, 5-shot MMLU, log-likelihood scoring), not by us. Model-specific evaluation-configuration errors, including prompt or formatting mismatches, cannot be ruled out and may affect the observed agreement measures. Near-chance accuracy (339 of 977 models below 0.30) does not by itself show a configuration error—many are small or weakly trained models—but it cannot exclude one; the S2 analyses restrict to models above 0.30.
- 7) **Null and population dependence.** Following [23], levels of excess agreement depend on the null model and on the models and items compared; we report a contrast that is stable across the nulls examined, but other populations or benchmarks may give different contrasts.
- 8) **Contamination.** If MMLU items appear in training data, agreement may partly reflect memorization shared by lineage.
- 9) **Temporal coverage of lineage pairs.** Pairs sharing a root were almost all released within a year of each other (primary: 711, 1,004, 259, 45 and 1 pairs in the 0, 1–2, 3–5, 6–11 and 12+ month bins). The data therefore cannot show whether the lineage association persists between fine-tunes released far apart; an exploratory lineage-by-gap interaction is not identified in the primary sample and is near zero, with wide intervals, in the expanded sample.
- 10) **Concentration.** The lineage contrast comes from 63 roots with two or more models in the primary sample (108 expanded); root-level inference accounts for this, but a few large roots carry substantial weight.

IX. REPRODUCIBILITY

All code, frozen rosters (with SHA-256 hashes), Hub metadata caches, validation reports and analysis outputs are in the project repository at <https://github.com/SudharsanSelvaraj/Identifiable-Variance-Decomposition-of-Correlated-Errors-in-Foundation-Models>. A single script (`scripts/run_exp04_final.py`) re-runs the twelve pre-registered analyses, the inference and outcome audits and the exploratory checks from the frozen data, verifies the roster hashes, and diffs every coefficient against the committed outputs (maximum absolute difference below 10^{-16}); all tables in Sections IV–VI are generated from those outputs by `scripts/make_exp04_tables.py`, and every figure, including the counts in the two

diagrams, by `scripts/make_exp04_figures.py`; Fig. 3 is the only illustration with hypothetical models. The leaderboard’s per-item details and metadata declare no licence, so neither the derived item-level answer files (103 MB) nor the copied metadata fields are redistributed; the full model lists are rebuilt from the public source by `scripts/rebuild_frozen_lists.py`, which verifies them against the SHA-256 hashes recorded when they were frozen, and the answer files are regenerated by `scripts/download_ollb_v1.py` from the public leaderboard and validated by `scripts/validation_report.py`. The pre-registration and its three dated amendments are in `docs/05_Experiments/Exp04_Leaderboard_Lineage_Era.md`.

X. CONCLUSION

With the few families and quarters that real model populations offer, per-model variance decomposition cannot estimate the relative contributions of lineage and release era precisely enough for a reliable comparison, even when the variance components are formally identified. Moving to pairs of models, with lineage taken from declared ancestry and release time measured in months, makes the question estimable. Across 977 validated open-weight models, a shared lineage root is associated with a 12–13 point higher chance of choosing the same wrong answer when both models err, robustly across twelve pre-registered analyses and the inference, outcome and control variants examined. The association holds against item-difficulty-aware null models. Beyond what the two models’ accuracies explain, release-month proximity shows no association of consistent sign with the choice of wrong answer; it shows a small positive association with which items fail together in the full samples, which does not persist among models scoring at least 30%. Within this population, lineage, not release timing, is the dominant correlate of shared wrong-answer choices.

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Use of AI-generated content. The authors used Claude (Anthropic; model Claude Opus 5.5, accessed through Claude Code) [26] in October 2026, with substantial use. It wrote and debugged the analysis, simulation, data-extraction and figure/table-generation code, and drafted the text of every section of this revised manuscript (Sections I–IX and Appendix A). The authors directed the study, approved each analysis decision and pre-registration amendment, reviewed and edited the text, and take full responsibility for the content; every reported number is produced by the released code from the committed outputs.

REFERENCES

- [1] E. Kim, A. Garg, K. Peng, and N. Garg, "Correlated errors in large language models," in *Proc. Int. Conf. Mach. Learn. (ICML)*, vol. 267, pp. 30038–30066, 2025.
- [2] J. L. Chen, "When does combining language models help? A co-failure ceiling on routing, voting, and mixture-of-agents across 67 frontier models," *arXiv preprint arXiv:2606.27288*, 2026.
- [3] S. Goel, J. Struber, I. A. Auzina, K. K. Chandra, P. Kumaraguru, D. Kiela, A. Prabhu, M. Bethge, and J. Geiping, "Great models think alike and this undermines AI oversight," *arXiv preprint arXiv:2502.04313*, 2025.
- [4] B. Laufer, H. Oderinwale, and J. Kleinberg, "Anatomy of a machine learning ecosystem: 2 million models on Hugging Face," *arXiv preprint arXiv:2508.06811*, 2025.
- [5] E. Horwitz, A. Shul, and Y. Hoshen, "Unsupervised model tree heritage recovery," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [6] N. Yax, P.-Y. Oudeyer, and S. Palminteri, "PhyloLM: Inferring the phylogeny of large language models," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2025.
- [7] Y. Li *et al.*, "Tracing the roots: A multi-agent framework for uncovering data lineage in post-training LLMs," in *Proc. 64th Annu. Meeting Assoc. Comput. Linguist. (Volume 1: Long Papers)*, 2026, pp. 9606–9625.
- [8] C. Kuai *et al.*, "A statistical framework for auditing behavioral dependence and induced bias in LLM judges," *arXiv preprint arXiv:2604.07650v2*, 2026.
- [9] H. D. Patterson and R. Thompson, "Recovery of inter-block information when block sizes are unequal," *Biometrika*, vol. 58, no. 3, pp. 545–554, 1971.
- [10] D. A. Harville, "Maximum likelihood approaches to variance component estimation and to related problems," *J. Amer. Statist. Assoc.*, vol. 72, no. 358, pp. 320–338, 1977.
- [11] S. R. Searle, G. Casella, and C. E. McCulloch, *Variance Components*. New York, NY, USA: Wiley, 1992.
- [12] R. L. Brennan, *Generalizability Theory*. New York, NY, USA: Springer, 2001.
- [13] S. Messing, "Hidden measurement error in LLM pipelines distorts annotation, evaluation, and benchmarking," *arXiv preprint arXiv:2604.11581*, 2026.
- [14] M. Fafchamps and F. Gubert, "The formation of risk sharing networks," *J. Develop. Econ.*, vol. 83, no. 2, pp. 326–350, 2007.
- [15] A. C. Cameron, J. B. Gelbach, and D. L. Miller, "Robust inference with multiway clustering," *J. Bus. Econ. Statist.*, vol. 29, no. 2, pp. 238–249, 2011.
- [16] P. M. Aronow, C. Samii, and V. A. Assenova, "Cluster-robust variance estimation for dyadic data," *Political Anal.*, vol. 23, no. 4, pp. 564–577, 2015.
- [17] B. Efron, *The Jackknife, the Bootstrap and Other Resampling Plans*. Philadelphia, PA, USA: SIAM, 1982.
- [18] T. G. Dietterich, "Ensemble methods in machine learning," in *Proc. Int. Workshop Multiple Classifier Syst.*, LNCS 1857, 2000, pp. 1–15.
- [19] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.
- [20] L. I. Kuncheva and C. J. Whitaker, "Measures of diversity in classifier ensembles and their relationship with the ensemble accuracy," *Mach. Learn.*, vol. 51, no. 2, pp. 181–207, 2003.
- [21] J. Kleinberg and M. Raghavan, "Algorithmic monoculture and social welfare," *Proc. Nat. Acad. Sci. USA*, vol. 118, no. 22, 2021.
- [22] R. Bommasani, K. A. Creel, A. Kumar, D. Jurafsky, and P. Liang, "Picking on the same person: Does algorithmic monoculture lead to outcome homogenization?" in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 35, 2022.
- [23] N. Jo, N. Garg, and M. Raghavan, "The subjectivity of monoculture," *arXiv preprint arXiv:2602.24086*, 2026.
- [24] D. Hendrycks *et al.*, "Measuring massive multitask language understanding," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2021.
- [25] E. Beeching *et al.*, "Open LLM Leaderboard (2023–2024)," Hugging Face, 2023. [Online]. Available: https://huggingface.co/spaces/open-llm-leaderboard-old/open_llm_leaderboard
- [26] Anthropic, "Claude (Claude Opus 5.5), accessed through Claude Code," large language model, Oct. 2026. [Online]. Available: <https://www.anthropic.com/claude>
- [27] L. Gao *et al.*, "A framework for few-shot language model evaluation," Zenodo, 2021, doi: 10.5281/zenodo.5371628.

APPENDIX A CHANGES FROM THE EARLIER VERSION

The earlier version of this manuscript (archived in the repository as `paper/src/archive/`) reported a sixteen-model evaluation and an identifiability gate. The following findings of that version are withdrawn or corrected.

- 1) **Gate.** Its gate tested the rank, condition number and variance inflation of a fixed-effects dummy design. For the random-effects model it fitted, the variance components were identified (covariance-basis rank 3/3); the condition number depended on the reference level (115–585 for the same fifteen-model design). The claim that the variance components were "not identifiable" is withdrawn; the correct conclusion, imprecision, is in Section III.
- 2) **Item-level outputs.** The evaluation code recorded only the first option's log-likelihood, so every per-question prediction was option A. The code is fixed and tested; those outputs are not used. The aggregate accuracy reported for Qwen-7B (0.2294) equalled the all-A rate and is withdrawn.
- 3) **Text errors.** Statements that a 30-model design "passes" at a condition-number threshold of 50 ($\kappa = 93$), that the design sweep enforced crossing and connectedness by construction (100 of 105 configurations were crossed and 60 connected), that a linear-probability model and a GLMM both reached the era boundary in 20–60% of replicates (only the GLMM did; the linear model 0–6.7%), and that a nested-design detector had "zero silent coverage" (the conditional quantity was undefined because no replicate went undetected) are corrected.
- 4) **Population history.** Four models in the sixteen-model study's pre-registered plan (Llama-3.1, Llama-3.3, Qwen1.5 and Phi-4-reasoning-vision-15B) were not evaluated, without a recorded reason; the two DeepSeek models were planned for imputation. That arm is no longer used for inference.

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